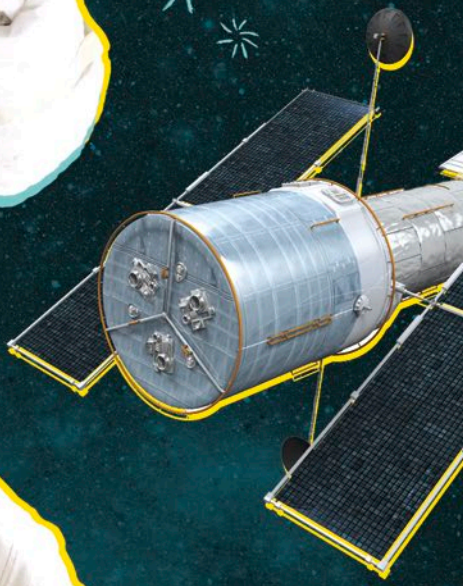
SPACE

FACTS OR FIBS





KRISTIN J. RUSSO

EDGE BOOKS



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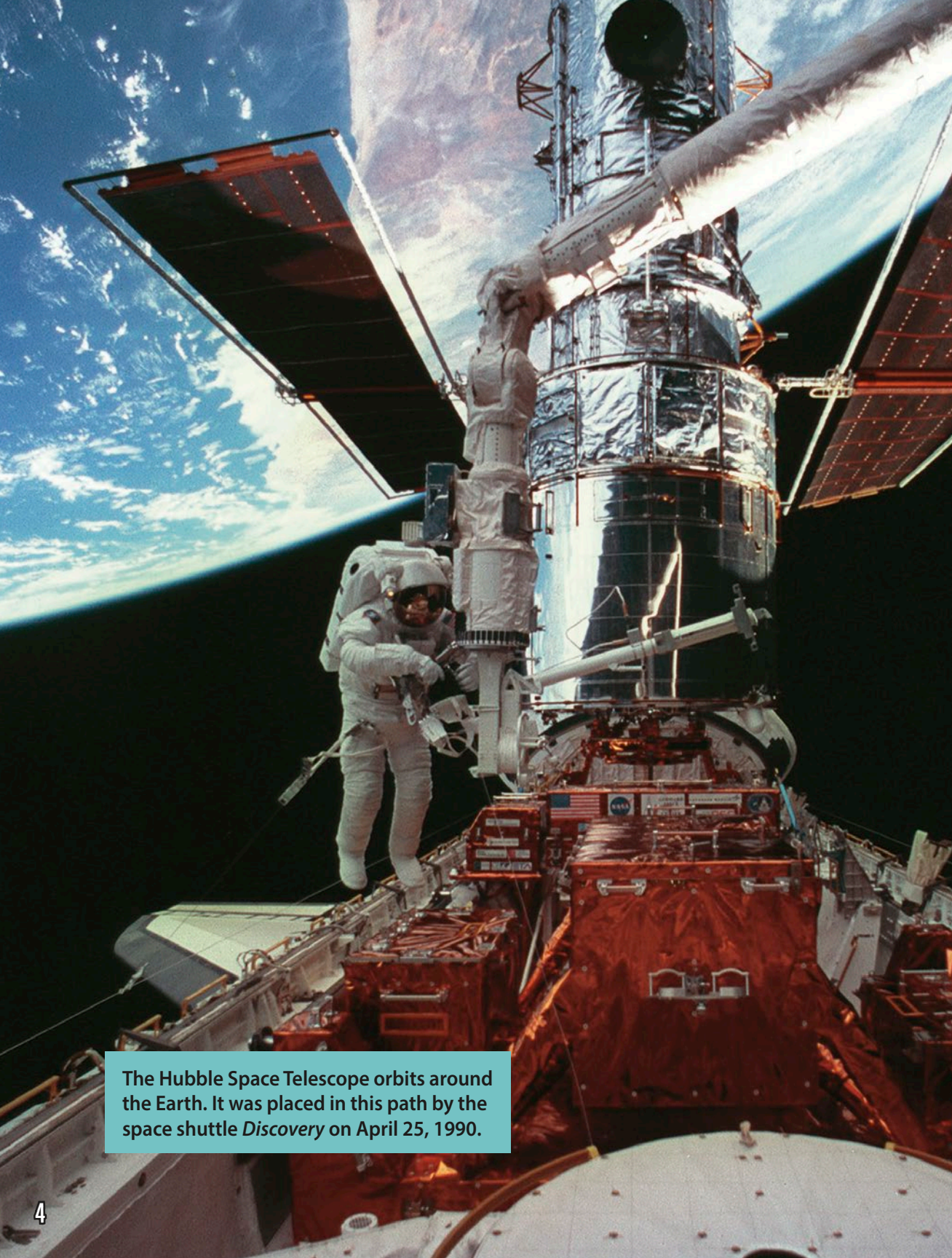
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The Hubble Space Telescope orbits around the Earth. It was placed in this path by the space shuttle *Discovery* on April 25, 1990.



INTRODUCTION

SPACE THROUGH TIME

Thousands of years ago, people made up stories to explain what they saw in the sky. Different groups had ideas about the sun, clouds, stars, and moon. Ancient Egyptians believed in Re, the sun god. He had children with special powers. These children became the planet Earth, the stars, the **atmosphere**, and the clouds. The Norse Vikings believed the sun and moon were pulled through the sky by chariots. The ancient Greeks believed in gods and goddesses. These beings controlled the sky. Later, the Greeks used math to explain how the planets and stars moved in the sky. Many of their ideas helped scientists make new discoveries.

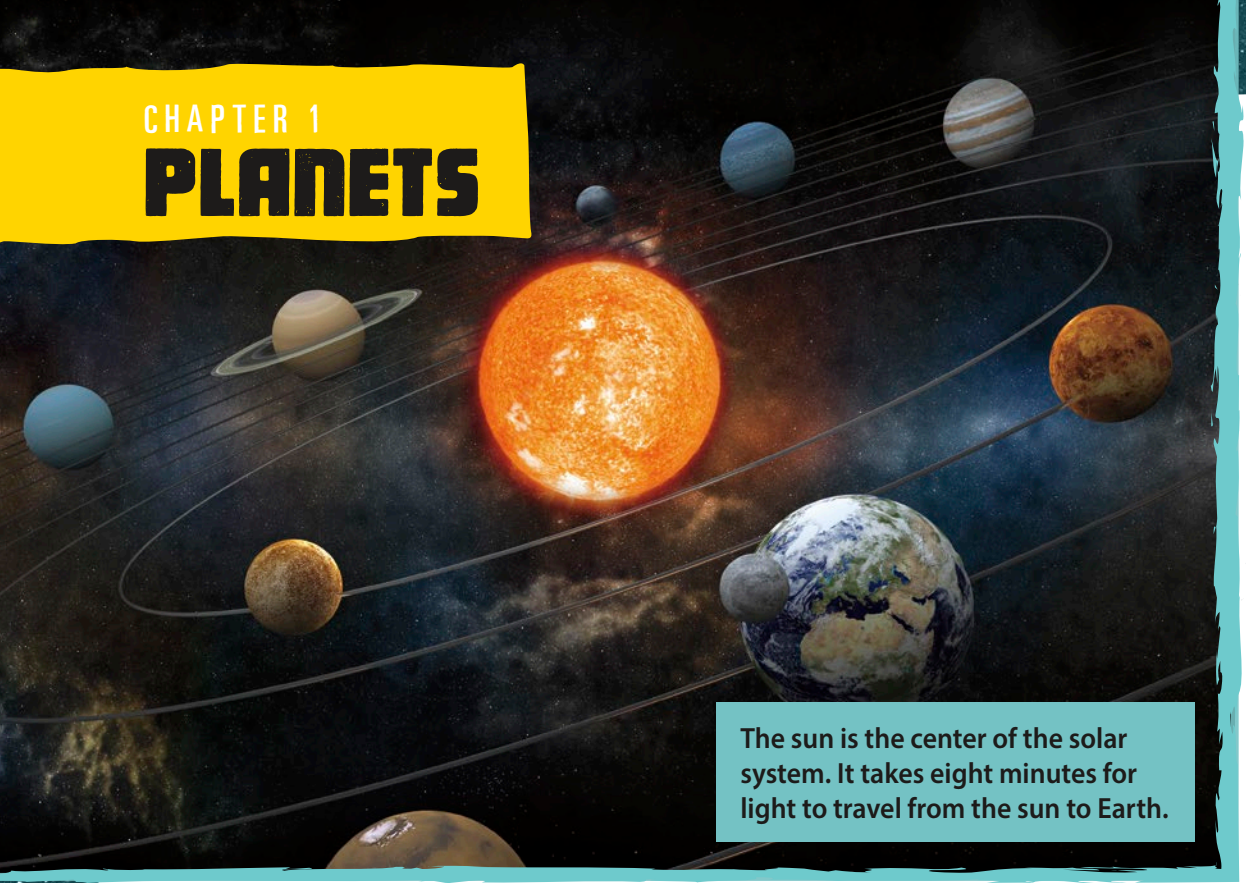
We no longer rely on myths and stories to help us understand space. Instead, we use scientific tools. Today astronauts and scientists work together to improve space exploration. They use telescopes and **satellites** to learn more than ever before about our **solar system**.

People used to believe Earth was flat, but now we know that is not true. People also used to think Earth was the center of the solar system. This is also not true. Today, scientists know a lot more about space than people did in the past. But they are still trying to figure out what is a fact and what is a fib when it comes to outer space.

atmosphere—the layer of gases that surrounds some planets, dwarf planets, and moons
satellite—a spacecraft that circles Earth; satellites gather and send information to Earth
solar system—the sun and the objects that move around it

CHAPTER 1

PLANETS



The sun is the center of the solar system. It takes eight minutes for light to travel from the sun to Earth.

People used to believe the sun circled around Earth. Scientists proved this to be false. Earth and all of the planets in our solar system travel around the sun. Scientists know that discovering a scientific fact can take time. For example, researchers are still trying to figure out how many planets exist in our solar system. They have watched the sky closely to learn how the solar system works. The solar system includes the sun and all of the planets that revolve around it.

IT'S TRUE! Mercury and Earth move at different speeds. When they move in different directions it is called "Mercury in retrograde." There is no scientific proof that this affects Earth. Some people believe that crazy things happen when Mercury is in retrograde. They blame accidents and communication mix-ups on the position of the planets.

FACT OR FIB?

There are nine planets in our solar system.

Evidence

There are several requirements that must be met before something can be considered a planet. First, a planet must travel around the sun. It must have a nearly round shape. The planet's gravity must be strong enough to push asteroids and other celestial bodies out of its path. It can also pull them in to make them part of the planet. For more than 70 years, scientists agreed that there were nine planets in the solar system.

Answer: UNDECIDED



In 2006 scientists decided that Pluto is not an official planet. It does not meet all the requirements. It is now called a "dwarf planet."

Today, there are eight official planets. They are Mercury, Venus, Earth, Mars, Jupiter, Saturn, Uranus, and Neptune. But there may be one more.

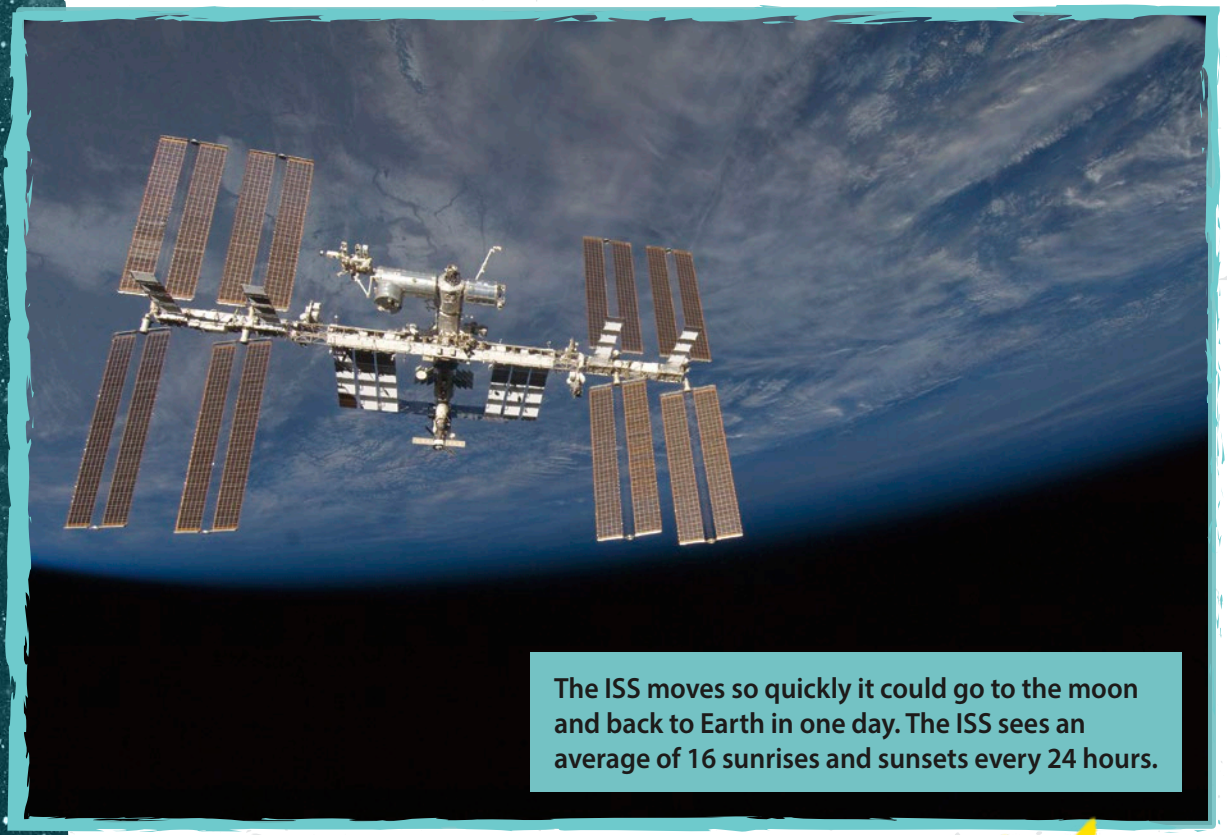
In January 2015, **astronomers** announced that they had discovered another possible planet. It is called Planet Nine. Planet Nine is far away from the sun. It would take about 10,000 to 20,000 years to complete its **orbit**. If scientists agree, Planet Nine could become an official planet. This would raise the number of planets in our solar system to nine.

astronomer—a scientist who studies stars, planets, and other objects in space

orbit—the path an object follows as it goes around the sun or a planet

FACT OR FIB?

Except for the sun and the moon, the International Space Station (ISS) is the brightest object in the sky.



The ISS moves so quickly it could go to the moon and back to Earth in one day. The ISS sees an average of 16 sunrises and sunsets every 24 hours.

The ISS is a large spacecraft. Astronauts can live there and work in space. Astronauts from many different countries share missions at the ISS.

International Space Station (ISS)—a large spacecraft where astronauts live and work in space

Evidence



There are large, gold-colored solar panels on the ISS. Solar panels use energy from the sun to generate electricity. The solar panels are powerful. They reflect so much light, they make the ISS appear as bright as Venus in the night sky.

Answer: FACT

In 2009 **NASA** attached new solar panels. These new panels mean that the ISS can harness even more energy from the sun. They also reflect more sunlight, enough to make the ISS the third-brightest object in the sky. It is also the brightest man-made object in the sky.

IT'S TRUE! The Hubble Space Telescope is named for Edwin Powell Hubble. He first discovered the existence of other galaxies. The telescope was launched into space in 1990. This strong telescope orbits Earth and gives scientists a detailed view of the universe.

NASA—a U.S. government agency that does research on flight and space exploration; NASA stands for National Aeronautics and Space Administration